

Centre Number	Candidate Number	Candidate Name
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NAMIBIA SENIOR SECONDARY CERTIFICATE

CHEMISTRY ORDINARY LEVEL

6117/2

PAPER 2 Structured Questions

1 hour 30 minutes

Marks 80

2023

Additional Materials: Non-programmable calculator
Ruler

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces provided on top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.
- You will lose marks if you do not show your working or if you do not use appropriate units.
- The Periodic Table is printed on page 14.

For Examiner's Use

1	
2	
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Total	

Marker	
Checker	

This document consists of **14** printed pages and **2** blank pages.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

1 Bromine has two stable isotopes, $^{79}_{35}\text{Br}$ and $^{81}_{35}\text{Br}$.

(a) State what 79 represents in $^{79}_{35}\text{Br}$.

..... [1]

(b) Define the term *isotope*.

..... [2]
.....
.....

(c) Complete Table 1.1 to show the number of protons, neutrons and electrons in the isotopes of bromine.

Table 1.1

particle	number of protons	number of neutrons	number of electrons
$^{79}_{35}\text{Br}$			
$^{81}_{35}\text{Br}$			

[2]

(d) Some element's isotopes are radioactive and have several uses.

(i) Describe **one** disadvantage of using radioactive isotopes as a source of energy.

..... [1]
.....

(ii) Give **one** use of carbon-14.

..... [1]
.....

[7]

2 Period 3 contains the elements sodium to argon.

Na Mg Al Si P S Cl Ar

(a) Give the element that has the largest atomic radius in Period 3.

..... [1]

(b) Sodium is a reactive metal.

(i) Describe **one** physical property of sodium.

..... [1]
.....

(ii) Write down the electronic configuration of the sodium ion.

..... [1]

(iii) Give **one** observation when sodium reacts with cold water.

..... [1]
.....

(iv) State the flame colour of sodium.

..... [1]
.....

(c) Magnesium reacts slowly with warm water to form a base, magnesium hydroxide.

(i) Define a *base* in terms of proton transfer.

..... [1]
.....

(ii) Suggest the pH of magnesium hydroxide and give the colour of universal indicator

pH..... [1]
colour of universal indicator [2]

(iii) Magnesium hydroxide can be used to treat indigestion by reacting with stomach acid. State the type of reaction this is.

..... [1]

(d) Aluminium combines with oxygen to form aluminum oxide.

Name the type of oxide aluminium oxide forms.

..... [1]
.....

(e) Silicon dioxide has a giant structure.

(i) Name the type of bonding in silicon dioxide.

..... [1]

(ii) Give **one** physical property of silicon dioxide.

..... [1]

(f) Phosphate, PO_4^{3-} is an oxyanion.

Deduce the formula of calcium phosphate.

..... [1]
.....

(g) Sulfur combines with carbon to give carbon disulfide, CS_2 .

Draw a 'dot-and-cross' diagram of the carbon disulfide molecule.

Show outer electrons only.

[2]

(h) Chlorine and compounds of chlorine are important in water treatment.

(i) Chlorine is added to water to make the water safe to drink.

Explain why adding chlorine makes water safe to drink.

..... [1]
.....

(ii) Describe a chemical test for chlorine.

test

result

..... [2]

(iii) Argon is a noble gas.

State and explain **one** use of argon.

use

explanation.....

..... [2]

[20]

- 3 Fig. 3.1 shows the electrolysis of molten lead(II) bromide.

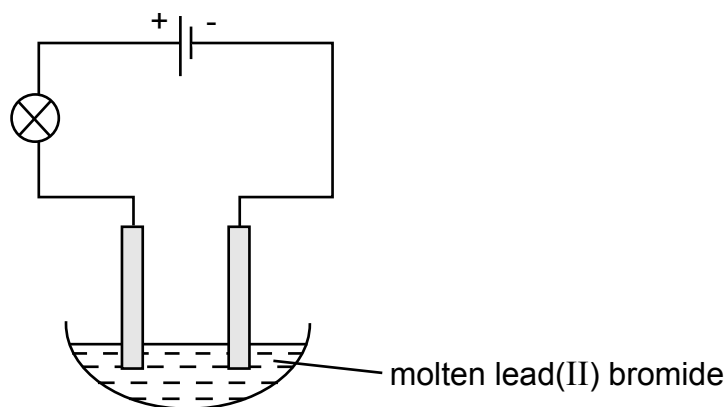


Fig. 3.1

- (a) Describe what is meant by the term *electrolysis*.

.....

.....

.....

.....

[2]

- (b) State the name given to the negative electrode.

.....

[1]

- (c) Give **two** observations when the circuit in Fig. 3.1 is closed.

1

.....

2

.....

[2]

- (d) Name the product at the positive electrode and give its half ionic equation.

name of product

half ionic equation

[2]

[7]

- 4 (a) When calcium carbonate is added to dilute hydrochloric acid, carbon dioxide gas is formed. A learner carried out an experiment to measure the volume of gas formed as the reaction proceeds. The student added a small mass of powdered calcium carbonate to an excess of 0.1 mol/dm^3 hydrochloric acid.
- Fig. 4.1 shows the graph of the results.

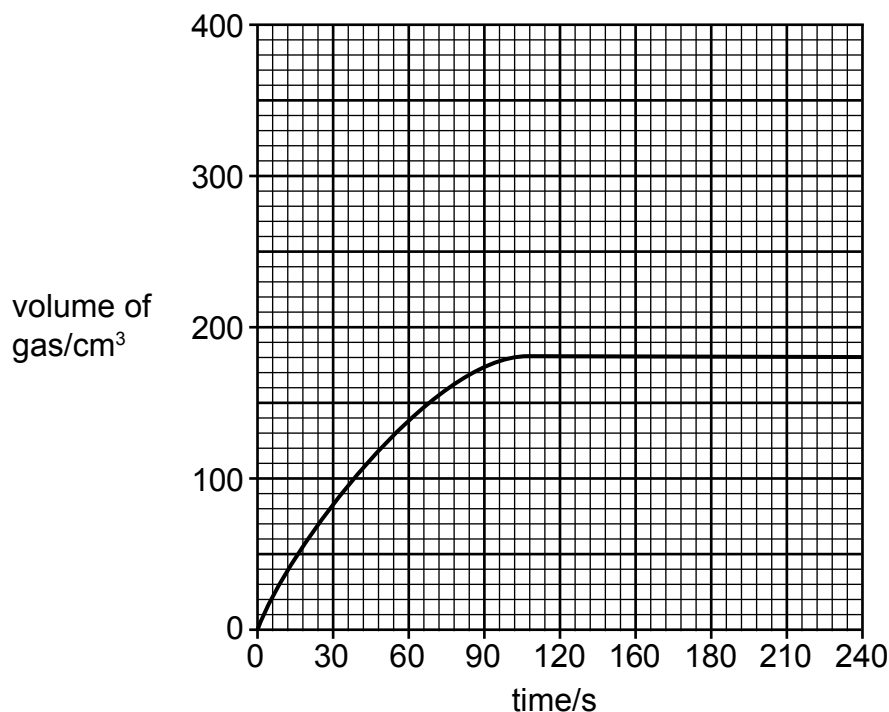


Fig. 4.1

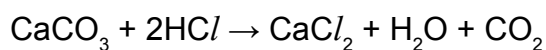
- (i) Name the **two** pieces of apparatus needed to take the measurements shown on the graph.

1

2 [2]

- (ii) The total volume of gas collected was 180 cm^3 at room temperature and pressure.

Calculate the mass, in grams, of calcium carbonate used.



mass = g [3]

- (b) (i) On Fig. 4.2, draw the graph you will expect if the same mass of calcium carbonate is added as large lumps instead of powder. All other conditions are the same as in 4(a).

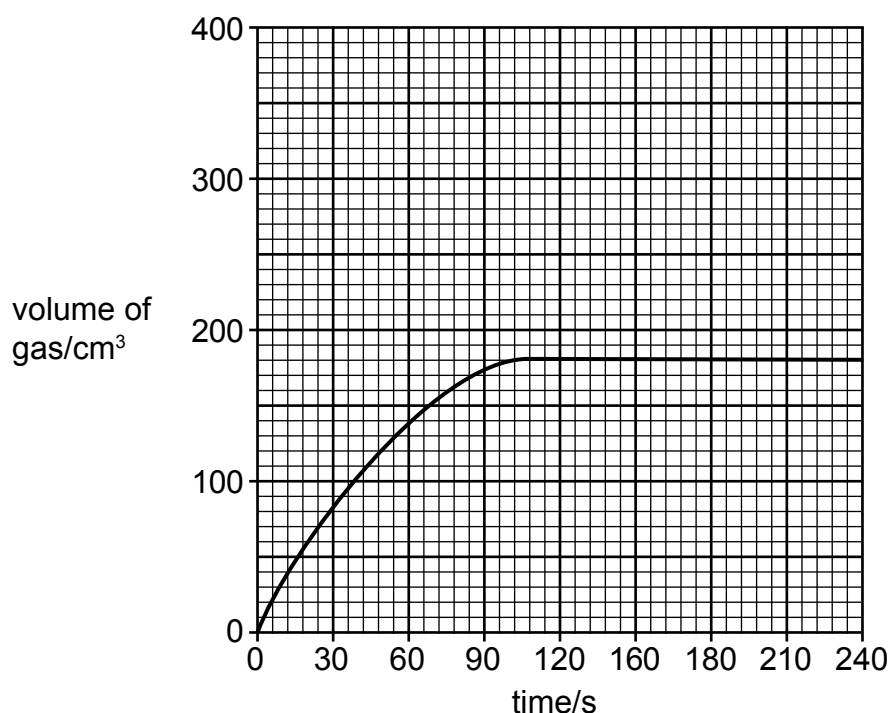


Fig. 4.2

- (ii) Explain why your graph in (b)(i) is different from the graph in Fig. 4.1.

.....

.....

.....

.....

- (c) State **one** other factor that affects the rate of a reaction.

.....

.....

[10]

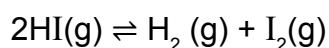
5 A good example of an endothermic reaction is dissolving a salt.

(a) Explain the meaning of the phrase *endothermic reaction*.

.....
.....

[1]

(b) Bond energy is the amount of energy, in kJ, that must be supplied to break one mole of a bond. Hydrogen iodide dissociates into its elements according to the equation below.



Bond	Bond energy in kJ/mol
H-H	436
I-I	151
H-I	299

Use the above data to calculate the overall energy change (ΔH) for the reaction shown.

energy change = kJ/mol

[3]

(c) The manufacture of sulfur trioxide is also an equilibrium reaction.



(i) The equilibrium is a dynamic equilibrium.

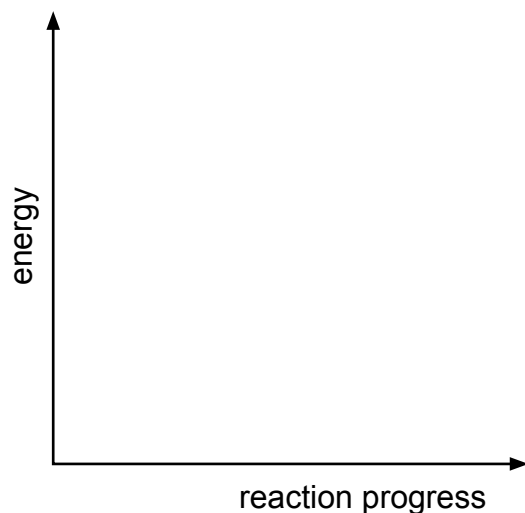
Explain what is meant by a *dynamic equilibrium*.

.....
.....

[1]

- (ii) Draw an energy diagram for the reaction. Use arrows on the diagram to show the activation energy, E_a , and the enthalpy change, ΔH .

Fully label your diagram. The axes are drawn for you.



[3]

- (iii) State the effects on the equilibrium reaction if there is:

an increase in the temperature,

.....

.....

[1]

an increase in pressure.

.....

.....

[1]

[10]

- 6 (a) Fig. 6.1 shows an experiment where zinc metal is immersed in a solution of copper(II) sulfate

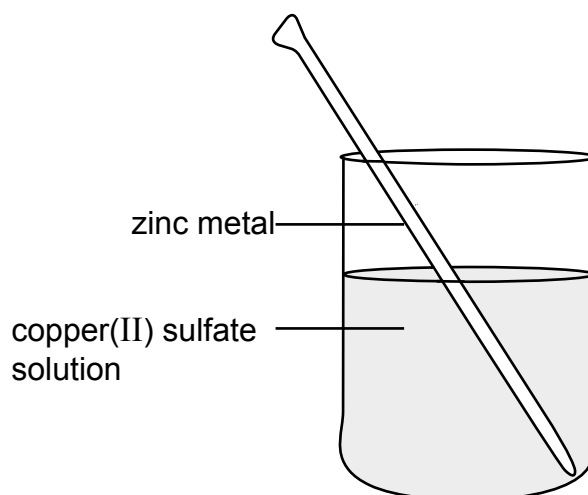


Fig. 6.1

- (i) Give **one** observation made when zinc reacts with copper(II) sulfate.

.....
 [1]

- (ii) Explain your observation in part (i).

.....
 [1]

- (b) Zinc is extracted from its ore.

- (i) Name the main ore of zinc.

..... [1]

- (ii) Zinc sulfide is converted to zinc oxide in an industrial process.

Describe how zinc sulfide is converted to zinc oxide in this industrial process.

.....
 [1]

- (iii) Zinc oxide is then reduced in a furnace.

Name the substance added to the furnace to reduce the zinc oxide.

..... [1]

- (c) Give **one** use of zinc metal.

..... [1]

[6]

7 Petroleum is a mixture of hydrocarbons.

Two of the processes carried out in an oil refinery are fractional distillation of petroleum and cracking of hydrocarbon fractions.

(a) Define the term *hydrocarbon*.

.....
 [2]

(b) (i) State **one** property of hydrocarbons that is used to separate petroleum into fractions.

.....
 [1]

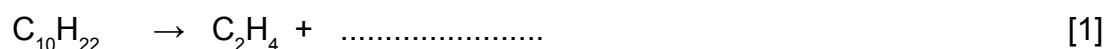
(ii) One of the fractions formed is naphtha.

Give **one** use of naphtha.

.....
 [1]

(c) The hydrocarbon, $C_{10}H_{22}$, can be cracked to make ethene and one other hydrocarbon.

(i) Complete the equation for this reaction.



(ii) Explain the meaning of the term *cracking*.

.....
 [1]

(d) Ethanol can be made from ethene.

Name the reagent and conditions needed to convert ethene to ethanol.

reagent
 condition..... [2]

(e) State the type of chemical change which occurs when ethanol is converted into ethanoic acid.

..... [1]

(f) State the name of the polymer formed from ethene.

..... [1]

(g) Polyesters such as terylene contain an ester linkage.

(i) Draw the structure of an ester linkage.

[1]

(ii) State the name of a natural polymer possessing an ester linkage.

.....

[1]

[12]

8 Ammonium sulfate is used in fertilisers.

(a) State the names of the **three** elements found in most fertilisers.

1

2

3

[3]

(b) Fertilisers containing ammonium salts are often slightly acidic.

State the name of a compound which farmers add to the soil to make it less acidic and explain why it is used.

name

explanation

.....

[2]

(c) Discuss the dangers of overuse of fertilisers.

.....

.....

.....

.....

.....

.....

[3]

[8]

DATA SHEET																				
The Periodic Table of the Elements																				
Group																				
I	II											III	IV	V	VI	VII	0			
												1 H Hydrogen 1							4 He Helium 2	
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10			
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18			
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	64 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36			
85 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54			
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	178 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	210 Rn Radon 86			
87 Fr Francium	226 Ra Radium	227 Ac Actinium																		
*58 - 71 Lanthanoid series																				
†90 - 103 Actinoid series																				
													162 Dy Dysprosium 66	165 Ho Holmium 67	167 Er Erbium 68	169 Tm Thulium 69	173 Yb Ytterbium 70	175 Lu Lutetium 71		
													98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium		
Key		a	X	a = relative atomic mass X = atomic symbol b = proton (atomic) number										b						

The volume of one mole of any gas is 24 dm³ at room temperature and pressure (r.t.p.).

a

X

b

a = relative atomic mass

X = atomic symbol

b = proton (atomic) number

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